



# An Approach for Designing Management Support Systems: The Design Science Research Process and its Outcomes

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## ABSTRACT

Design science research involves creating and evaluating innovative methods and approaches to be used in design practice. We present an approach to be used in the process of designing Management Support Systems (MSS). The nature of managerial work makes the design, development, and implementation of MSS a major challenge. The MSS literature suggests that determining MSS requirements and specification of MSS are the most critical phases in MSS design and development. We present an approach that can be used as a guide for MSS design, with a primary focus on MSS requirements determination and how requirements can be fulfilled using information and communication technologies (ICT). The approach builds on Quinn and associates' competing values model (CVM) of organizational effectiveness. The approach can guide MSS designers in designing MSS that support different managerial roles, i.e., the development of MSS that support managerial cognition, decision, and action.

## Keywords

Design science, design approach, management support systems

## 1. INTRODUCTION

Design science research involves creating and evaluating innovative methods and approaches to be used in design practice. We present an approach to be used in the process of designing Management Support Systems (MSS). MSS is a major

Information Systems (IS) class [17]. Many types of computer-based IS have been developed to support managers, including Decision Support Systems (DSS), Executive Information Systems (EIS), Group Support Systems (GSS), Knowledge Management Systems (KMS), and Business Intelligence (BI) systems [17, 20]. MSS usually refer to systems with a broader set of capabilities than, for example, EIS and DSS. An MSS can include electronic communications, modeling capabilities and organizing tools. In this paper we use the term Management Support Systems to denote ICT-enabled IS that are supposed to support managers, i.e. support managerial cognition, decision, and action.

A major problem in MSS design and development is requirements specification [35]. One reason for the problem is that managers perform quite a lot of different roles and activities [50, 51] and managers' work is not easy to model using traditional modeling techniques, like process modeling techniques. We present an approach that can be used to guide MSS design, with a primary focus on MSS requirements determination and how requirements can be fulfilled using information and communication technologies (ICT). The approach builds on Quinn and associates' competing values model (CVM) of organizational effectiveness. The presented approach can guide MSS designers in designing MSS that support different managerial roles, i.e. the design of MSS that support managerial cognition and behavior.

As mentioned above, studies suggest that a major problem in MSS design and development is requirements specification [73]. Studies also show that a legitimate need is a key to MSS success [17, 29, 35, 73]. This paper addresses the issues and problems in MSS design in a novel way. In doing so, we build our work on three postulates.

First, MSS is not a particular technology in a restricted sense, but primarily a perspective (vision) on managers and managerial work, the role of ICT and ICT-enabled MSS and how to realize this vision in practice. There is room for different views on MSS and obviously also room for different MSS design approaches and methods. But as noted by Walls et al. [71] and Hartono et al. [35]

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there exists little theoretical work that directly guides MSS design. Our MSS design approach is based on a well-established theory and model: the competing values model [59].

Second, it is a misconception to think of MSS as systems that just provide managers with information. Part of the misconception lies in the use of the concept information, for example, in Executive Information Systems. MSS are systems that do more than provide information. MSS are systems that should support managerial cognition, decision, and action—providing information is only one of several means.

Third, effectiveness is a critical construct in IS and can not be ignored in theory and research. Improved effectiveness is often claimed as a desired end for many ICT-enabled IS. Our MSS design approach is based on the CVM which is underpinned by an effectiveness perspective.

The remainder of the paper is organized as follows. The next section discusses approaches and methods for the design and development of MSS. This is followed by a presentation of the CVM and how it can be an underpinning theory and model for an MSS design approach. The following section presents our MSS design approach and how it can be used in MSS design. The final section presents conclusions and discusses further research and development of the approach.

## 2. MSS DESIGN APPROACHES AND METHODS

There exist a large number of different approaches, methodologies, and methods for the design and development of ICT-based information systems [2]. More than 15 years ago, Jayaratna [39] estimated that there were more than 1000 ‘brand name’ approaches worldwide. Fitzgerald [29] and Watson et al. [73] point out that there, for good reasons, are differences between traditional systems development approaches and MSS development approaches. After reviewing a number of IS development approaches, Fitzgerald [28] suggests that new approaches are needed and that they can be based on, for example, current management and organizational theories. We present an MSS design approach based on a current management theory and model.

An issue in the design and development of MSS is related to the relationships between designers, users (managers), and other stakeholders. In general, the writings on IS development recommend that users should be involved in the design process. A problem often encountered in MSS development is that the users have very limited time for participating in the development process [27, 29, 52, 73]. A way out of this problem is to develop MSS design approaches and methods that are based on contemporary management models and theories. Such approaches can support MSS designers in their design work and enhance communication with the users/managers. Before presenting the underpinning theory and model of our MSS design approach, we review some of the current MSS design approaches and models.

In a study focusing on the MSS approaches and methods used by organizations in the US, Watson et al. [73] found that only two formal methods were used, namely: the critical success factors method (CSF) and the strategic business objectives (SBO) method.

The critical success factors (CSF) approach can be used to identify managers’ information needs [60]. Critical success factors are “...the few key areas of activity in which favorable results are absolutely necessary for a particular manager to reach his goals” [4]. The approach has clear links to writings in strategic management. The CSF method was extended to include critical decisions and critical assumptions [36]. Critical decisions (CD) are the decisions or decision processes that directly affect the success or failure of the CSFs. Critical assumptions (CA) are the assumptions that lead a manager to believe that the CSFs are valid. An MSS should provide information and support related to the CSF, CD, and CA. Lu and Wu [44] presented the IDEAL method for MSS development. The method builds on CSF and critical success actions (CSA). CSA concern the decisions of actions, i.e. what to do if a particular CSF goes wrong.

Volonino & Watson [70] proposed the strategic business objectives (SBO) methodology. The SBO methodology derives managers’ information needs from an organization’s objectives and the organization’s critical business processes. SBO consists of six steps; the early ones being: identify an organization’s strategic business objectives; identify business processes that are critical to the business objectives; prioritize strategic business objectives and critical business objectives; define information needed to support the critical business processes.

Walls et al. proposed a theory for designing vigilant EIS. A vigilant EIS is an information system that helps “...an executive [manager] remain alertly watchful for weak signals and discontinuities in the organizational environment relevant to emerging strategic threats and opportunities.” [71]. Walls et al. also described an information requirement determination method called the “Critical Attention Tracker.”

The above approaches and methods build in part on management and executive literature. They focus primarily on identifying information needs and especially critical/key performance indicators. Although, they can be useful, they have one major limitation. Since, they to a large extent focus on information needs they are not complete in generating suggestions for use of ICT for supporting managers. We suggest that focus should be on managerial roles and an MSS design approach should address how the managerial roles can be supported by ICT.

The approach we took in developing our MSS design approach was to review some of the descriptive and prescriptive management literature. The assumption was that the review should point to areas in which ICT can logically aid managers and that it should be possible to develop an MSS design approach based on the literature. Rockart and De Long [61] recommended such an approach. The theory and model underpinning our approach is the work of Robert Quinn and associates. The main reasons for using their work were that:

- They present a comprehensive framework/model of management work and in their work there is a strong link between theory and empirical studies.
- They address the link between how managers perform their managerial roles and performance (effectiveness).
- Their framework/model can be used to understand how ICT can be used to support managers.

The next section presents the competing values model (CVM).

### 3. THE COMPETING VALUES MODEL AND MSS DESIGN

Organizational effectiveness is one of the foundations of management and organization theory, research, and practice [6, 8, 43]. The competing values model (CVM) was, in part, developed to clarify the effectiveness construct [54, 57, 58]. The CVM perceives organizations as paradoxical [5, 6, 56], and it suggests that to achieve high performance requires an organization and its managers to simultaneously perform paradoxical and contradictory roles and capabilities [6, 34]. The CVM of organizational effectiveness incorporates three fundamental paradoxes acknowledged in the literature: flexibility and spontaneity vs. stability and predictability (related to organizational structure); internal vs. external (related to organizational focus); and means vs. ends [54, 58].

Quinn and Rohrbaugh [58, 62] found that most measures of effectiveness reflect one of four organizational models: internal process model (IP), rational goal model (RG), open systems model (OS), or human relations model (HR). The four models provide competing views on the meaning of organizational effectiveness. The human relations model is characterized by a focus on internal flexibility to develop employee cohesion and morale. It stresses human resource development, participation, and empowerment. The internal process model is characterized by a focus on internal control and uses information management, information processing, and communication to develop stability and control. The rational goal model is characterized by a focus on external control and relies on planning and goal setting to gain productivity and accomplishment. The open systems model is characterized by a focus on external flexibility and relies on readiness and flexibility to gain growth, resource acquisition, and external support.

The CVM points out the simultaneous opposition in the criteria that organizational members use to judge effectiveness. An organization does not pursue a single set of criteria. Instead an organization pursues competing, or paradoxical, criteria simultaneously. Organizations are more or less good in pursuing the criteria, and, according to the CVM, organizations differ in their effectiveness [22].

Quinn [54, 59] translated the construct of effectiveness into managerial roles—two for each of the four organizational models. In the monitor role (IP) a manager collects and distributes information (mainly internal and quantitative information), checks performance using traditional measures, and provides a sense of stability and continuity. In the coordinator role (IP) a manager maintains structure and flow of the systems, schedules, organizes and coordinates activities (logistic issues), solve house keeping issues, and sees that standards, goals and objectives, and rules are met.

In the director role (RG) a manager clarifies expectations, goals and purposes through planning and goal setting, defines problems, establishes goals, generates and evaluates alternatives, generates rules and policies, and evaluates performances. In the producer role (RG) a manager emphasizes performance, motivates members to accomplish stated goals, gives feedback to members, and is engaged in and supports the action phase of decision making.

In the innovator role (OS) a manager interacts with the environment, monitors the external environment (environmental scanning), identifies important trends, is engaged in business and competitive intelligence, develops mental models, convinces others about what is necessary and desirable, facilitates change, and shares "image and mental models." In the broker role (OS) a manager obtains external resources, is engaged in external communication, tries to influence the environment, and maintains the unit's external legitimacy through the development, scanning, and maintenance of a network of external contacts.

In the facilitator role (HR) a manager fosters collective efforts, tries to build cohesion and teamwork—building the "trustful organization", facilitates participation and group problem solving and decision making, pursues "moral" commitment, and is engaged in conflict management. In the mentor role (HR) a manager is engaged in the development of employees by listening and being supportive, is engaged in the development of individual plans, and gives feedback for individual and team development.

After Quinn and Rohrbaugh's initial studies, research on CVM has proceeded. This research shows that the CVM has utility as a general framework for organization and management research and practice. The CVM has also been used in the IS field—see, for example, Carlsson and Widmeyer [15, 16], Sääksjärvi and Talvinen [67], Järvinen [40], and Carlsson and Leidner [13, 14].

Research suggests that effective managers are capable of balancing and performing contradictory and complex roles [33, 34]. Some studies have tried to explain why some managers are considered more successful than others [54]. Research suggests that it is possible to link management behavior to firm performance; for example, Hart and Quinn's [34] study suggests that managers having the ability to play multiple and competing roles produce better firm performance, especially with respect to organizational growth and innovation (business performance) and organizational (stakeholder) effectiveness. Denison et al. [22], in a contingency-based study, empirically tested the CVM and the associated roles. They found support for the model and the roles, especially for managers that were considered high performing. Denison et al.'s study led them to define effective leadership as "...the ability to perform the multiple roles and behaviors that circumscribe the requisite variety implied by an organizational or environmental context." [22]—the notion of requisite variety is adapted from Ashby [1].

Based on the CVM and Denison et al.'s definition of effective leadership, we define an MSS to be effective to the extent that it is used by a manager in such a way as to support the manager in his different managerial roles, and support managerial cognition and behavior that circumscribe the requisite variety implied by the organizational and environmental context. Our definition of effective MSS is based on a contingency view and postulates: 1) there is no best MSS, and 2) any MSS is not equally effective. The first postulate means that we can not say that a specific MSS, with certain capabilities and characteristics, is the best MSS in all situations. The second postulate means that in a specific situation all MSS are not equally effective. It is possible to differentiate between MSS in terms of their effectiveness.

Hence, the goal of our approach is to be a guide in MSS design and should lead to the development of MSS which when used will lead to increased effectiveness—according to the definition of an effective MSS.

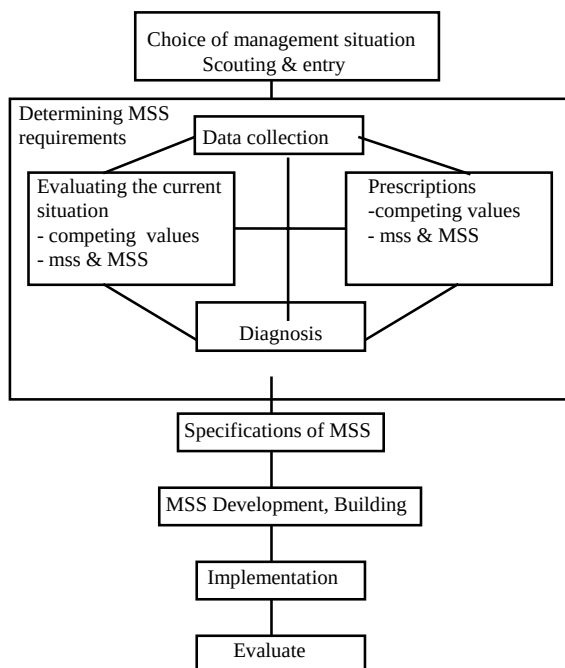
## 4. A NEW MSS DESIGN APPROACH

Before presenting the MSS design approach a few things about the approach have to be clarified. Watson et al.'s [73] study (discussed above) showed that in most cases a number of different "non-formal" approaches, methods, and techniques were used along with formal approaches in MSS development. Non-formal methods include, for example, discussions with support personnel, attending meetings in order to enhance the understanding of what information managers need. This mix of formal and non-formal approaches, methods, and techniques was also noted by Fitzgerald [29]. Avison and Fitzgerald [2] and Fitzgerald [28] point out that the search for the "grand"—and rigid—design approaches and methods might be over, and the best we can do in some cases is to develop approaches and methods that can be used as guides and to develop approaches suitable for specific types of systems.

Based on the above, our MSS design approach should primarily be perceived as an approach for guiding MSS design and in most cases the approach has to be supplemented with other formal and non-formal approaches, methods, and techniques.

### 4.1 The MSS Design Approach

Our CVM-based MSS design approach is depicted in Figure 1. The approach starts with the choice of management situation, scouting and entry phase. It includes how to set up the MSS design project and finding sponsors and champions. The importance of this phase is stressed in the MSS literature [17], but since there exists a good body of knowledge on these issues we will not address the phase here—the reader is commended to consult Watson et al. [73], Rockart & De Long [61], Fitzgerald [29], Fitzgerald & Murphy [30], and Clark et al. [17] on these issues.



**Figure 1. MSS design approach**

The second phase (determining MSS requirements) consists of data collection, evaluating the current situation, diagnosis, and

prescriptions. The third phase includes the technical specification of the MSS, which is followed by the building phase. We, as well as other MSS writers, have found it useful to use prototypes and to develop MSS iteratively. The fifth phase (implementation) includes implementing the MSS, i.e. putting the MSS in the hands of the managers. It also includes education/training and especially for new users a hot-line (in some cases available on a 24 hours basis). The final phase is evaluation, which in most cases brings us back to the second phase. Although, the figure shows a straightforward process, there are in most cases iterations between the different phases.

In the next sections we focus on the second and third phase, since these are the strengths of our approach and also probably the most critical phases in MSS development [72, 73, 74].

### 4.2 Determining MSS Requirements

This phase consists of four activities. The first activity is collection of data in order to: 1) evaluate the current situation, 2) be able to diagnose, and 3) prescribe desired changes.

#### 4.2.1 Evaluating the current situation

Evaluating the current situation includes: 1) evaluating the competing values and what managerial roles are performed (played), and 2) evaluating the mss (non-computer based systems) used, and if a MSS is used evaluating the use of the system.

In order to evaluate the current situation, different instruments developed and tested by Quinn and associates can be used [7, 54, 59]. By using the instrument it is possible to evaluate what different managers perceive as the effectiveness constructs, which roles they perceive as critical, and how much efforts they are putting into the different roles. In using the MSS design approach, we have, for example, used the "competing values leadership instrument: self-assessment" [54]. This instrument captures a manager's perception of what roles he is playing and to what degree. We have also used the "competing values leadership instrument: extended version" [54]. This instrument consists of 32 questions (behaviors) and a person completing the instrument has to, on a 7-point scale, respond to how frequently the person performs a specific behavior today and how often the behavior should be performed. The "competing values organizational effectiveness instrument" [54] has also been used—this instrument measures perceptions of organizational performance. The results can be presented for individual managers or as a summary of individual managers' perceptions.

In part, due to the problem of involving users (managers) in the design process, we have developed a supplementary way to identify the requirements for managerial behavior and cognition. Following the definitions of effective management and effective MSS, requirements for managerial behavior and cognition can be "derived" from an organization's external and organizational context. Carlsson and Leidner [14] present three contextual characteristics that can be used to identify requirements for managerial behavior and cognition: organizational environment, organizational strategy, and organizational structure. They use the following characteristics: 1) for the external environment: turbulence, competitiveness, and complexity—adapted from Huber et al. [38], 2) for the strategy: prospectors, defenders, analyzers, and reactors strategy—adapted from Miles and Snow [49], and 3) for the organizational context: centralization of decision making, standardization of procedures, specialization of

functions, and interdependence of organizational units and processes—adapted from Huber et al. [38]. The model presented by Carlsson and Leidner has been used to identify the requirements for managerial variety.

Using the CVM presented in Section 3, it is possible to generate MSS characteristics and capabilities. These are presented in Figure 2. The first subtype (MSS-IP) supports the internal process model and the associated managerial roles; it has an internal and control emphasis. The ends for MSS-IP are stability and control. Most functional and cross-functional quantitative computer-based IS (CBIS) can be used as MSS-IP. Traditional accounting information systems and production systems are good examples of systems used for supporting the IP-model. In most cases the MSS-IP can be built “on top of” transaction-based CBIS (operational systems), for example as a portal on top of an Enterprise Resource Planning (ERP) system [12]. Studies suggest that many MSS are built on top of existing internal CBIS [73] and most of these are closely associated with the IP model [12]. From a manager’s point of view the performance objective of MSS-IP is to provide user-friendly support for control and monitoring processes.

| Human Relations Model                                |  | Open Systems Model                              |  |
|------------------------------------------------------|--|-------------------------------------------------|--|
|                                                      |  | Structure                                       |  |
|                                                      |  | Flexibility                                     |  |
| Ends: Value of human resources                       |  | Ends: Resources acquisition,                    |  |
| Means: Cohesion, Moral                               |  | External support                                |  |
|                                                      |  | Means: Flexibility, Readiness                   |  |
| Systems characteristics and capabilities:            |  | Systems characteristics and capabilities:       |  |
| Communication & conferences (CSCW & Web 2.0)         |  | Environmental scanning & filtering (vigorantly) |  |
|                                                      |  | Interorganizational linkages                    |  |
| Internal                                             |  | External Focus                                  |  |
| Ends: Stability, Control                             |  | Ends: Productivity, Efficiency                  |  |
| Means: Information management, Communication         |  | Means: Planning, Goal setting, Evaluation       |  |
| Systems characteristics and capabilities:            |  | Systems characteristics and capabilities:       |  |
| Monitoring & Controlling, Record keeping, Optimizing |  | Modeling, Simulation, Forecasting               |  |
|                                                      |  | Control                                         |  |
| Internal Process Model                               |  | Rational Goal Model                             |  |

**Figure 2. Competing values model and systems characteristics and capabilities.**

The second subtype (MSS-RG) has an external and control emphasis. MSS-RG should support a manager in handling semi-structured problems. Examples of MSS-RG capabilities and features are what can be found in “traditional” Decision Support Systems (DSS), i.e. support for goal setting, forecasting, simulations, and sensitivity analyses. Although many DSS have an individual focus, DSS can also be group-oriented (GDSS) and support management teams in strategic planning processes.

The third subtype (MSS-OS) has an external focus and an

emphasis on structural flexibility. MSS-OS supports a manager in identifying problems and possibilities by support for environmental scanning, issue tracking, and issue probing. Environmental scanning can be quantitative or qualitative oriented and can include: industry and economic trends, legislative issues, competitor activities, new product and process development, patents, mergers and acquisitions, alliances, national and international events.

The MSS-HR subsystem helps an organization and its managers in the development of the human capital of the organization. Traditionally, MSS-HR capabilities and features of importance have been similar to what can be found in Computer Supported Cooperative Work (CSCW) systems and groupware; for example, ICT, like e-mail, voice mail, and videoconferencing can be used in MSS-HR to overcome distance and time. The rising popularity of user-driven online services, for example, Wikipedia, has drawn attention to a group of technological developments known as Web 2.0 or Enterprise 2.0 [46, 63, 68]. The technologies include, for example, blogs, Wikis, podcasts, and peer-to-peer networking and are promising to change inter- and intra-organizational collaboration and communication [47, 48].

The above has been used in developing a number of instruments that can be used to evaluate mss and MSS use—the instruments are ideal instruments that have to be adapted to a specific context. The instruments—16-32 questions and 5/7-point scales—are used to have individuals to evaluate how frequently they use mss/MSS for performing specific behaviors (managerial roles) and the person’s perceived value of the support (the mss/MSS). We have also in some cases asked the respondents to answer questions about an ideal situation.

Using the instruments and other in-formal methods and techniques it is possible to evaluate the current use of mss and MSS and their perceived usefulness.

#### 4.2.2 Diagnosis and desired changes (prescriptions)

A good fit between the current situation and the desired situation (i.e. the managers see no need for a change and the support they receive is good) means that there is no need for a new MSS. But, it is still possible to discuss the design of an MSS, but the primary purpose of the MSS would be to improve the efficiency—the MSS will primarily reinforce and improve the current state (primarily MSS as a personal tool).

If there is a misfit between the current situation and the desired situation or there is a misfit between current support and desired support, then there is a possibility to develop an MSS. In this case the MSS will be used as a means (tool) for focusing organizational attention and learning as well as a means for organizational change [10, 14, 69, 73].

The result of the diagnosis phase will be recommendations concerning how the competing values should be changed and how an MSS should support the different managerial roles in the future. The result will be used in the next phase.

### 4.3 MSS Specification

The output of phase two is desired changes. In the “Specification of MSS” phase these will be transformed into MSS specification; that is, taking the requirements and specify MSS characteristics and capabilities. This will later on lead to the development of an

MSS and hopefully when managers use it will lead to that the desired changes are fulfilled.

We now discuss how MSS can support the different managerial roles by looking at MSS capabilities and MSS information content. Although we propose how MSS can support different managerial roles it is not an MSS's capabilities and information characteristics that will determine the system's effectiveness, but how the capabilities are actually used by managers in their different roles. It should also be noted that new ICT that can be used for MSS will constantly hit the market and technologies will continue to change and evolve rapidly. (See Power's DSS-page at URL <http://dssresources.com> for reviews of the changing technological context of DSS and MSS.)

To discuss the usefulness of ICT for MSS we use an ICT typology. There are several different ways to classify ICT [31, 32]. We have adapted the classification presented by George et al. [32]. They describe seven different ICT. There are no clear borders between the different types of technologies and a specific MSS has features from more than one type of ICT.

The first type is communication technologies, where ICT is used to foster and support team, organization, and inter-organizational communication. The second type is coordination technologies, which is used to coordinate resources, projects, people, and facilities; for example to ensure that there is a synchronization of tasks and activities both horizontally and vertically. The third type is filtering technologies which are used to filter and summarize information. The fourth type is decision making technologies and techniques. This type is used to improve the effectiveness and efficiency of decision making processes. The fifth type is monitoring technologies, which is used to monitor the status of organizational activities and processes, industry trends, etc. The sixth type is data/knowledge representation technologies, which is used to represent and store data, text, images, animations, sound, and video. The seventh type is processing and presentation technologies which are ICT used to process data and present information. Table 1 suggests the extent to which each of the seven ICT can be useful in building the different MSS subsystems.

For MSS-IP the key capabilities are: 1) monitoring the status of organizational activities and processes, 2) filtering and summarizing critical information, and 3) support for coordination of resources, projects, people, and facilities.

Monitoring support can be provided by using "standard" MSS software to build MSS for status access and exception reporting and enhanced with structured or free wheeling drill down capabilities. MSS can also include the use of data warehouses, multidimensional databases, and OLAP (On Line Analytical Processing) which, for example, can give a manager the ability to slice and dice a multidimensional database (datacube). The Web browser is becoming an almost universal interface. The browser can be used to integrate a MSS with the organization's intranets. An increasing number of MSS software packages are developed to be Web-ready. This means that OLAP on data from a data warehouse or a multidimensional database can be directly accessed via the organization's intranet or through a portal [12]. In general, these changed and new capabilities make it possible to use traditional internal control systems more actively and in an ad hoc manner.

**Table 1. ICT technologies for the four MSS subsystems.**

| Building components \ MSS subsystem | Model: | Internal process | Rational Goal | Open Systems | Human Relations |
|-------------------------------------|--------|------------------|---------------|--------------|-----------------|
|                                     | Type:  | MSS-IP           | MSS-RG        | MSS-OS       | MSS-HR          |
| Communication                       |        | **               | **            | **           | ***             |
| Coordination                        |        | ***              | **            | *            | ***             |
| Filtering                           |        | ***              | **            | ***          | *               |
| Decision making                     |        | **               | ***           | *            | **              |
| Monitoring                          |        | ***              | **            | ***          | **              |
| Data/knowledge representation       |        | **               | ***           | **           | *               |
| Processing & presentation           |        | **               | ***           | **           | **              |

Key requirement: \*\*\*

Somewhat useful: \*\*

Little use: \*

Filtering support can be provided by using technologies for filtering and summarizing information from internal information sources. If an organization uses a data warehouse or data marts filtering can be done in the "extract, transfer, load" process and in the "analysis and presentation" process. In order to enhance this, software agents and push technology can be used.

Coordination support, for example, to ensure that there is a synchronization of tasks and activities both horizontally and vertically, can be provided by project management tools, electronic calendars, and workflow management systems. It is possible to use information generated in a workflow management systems to assess the workflow.

Managers in organizations working in increasingly turbulent environment are likely to increase their use of internal real-time information [3, 11, 26]. Traditional MSS are often based on financial data, but many organizations are rethinking their performance measures [23]. There is a shift from treating financial figures as the foundation for performance measurement to treating them as one among a broader set of measures [41]. Performance measures related to quality, customers, learning and growing, and even intellectual capital are increasingly used by companies. Balanced Scorecard (BSC) software has been launched by a number of companies and BSC will probably be a standard feature of MSS software and ERP systems. ERP systems can in part fulfill new information requirements. Alternatively, ERP systems make it possible to in a simple way pipeline data from the enterprise system to, for example, a data warehouse that are used by an MSS (Although it from a technical perspective might be simple to pipeline the data, designing an adequate data infrastructure for a data warehouse is far from a simple task [42].) In many cases this trend requires an organization's top-managers to become more actively involved in information ecology [18, 21]. A growing area for presenting relevant information to managers is the use of performance dashboards [24] often linked to organizational business performance management (BPM).

For the MSS-RG three capabilities are keys: 1) support for improving the effectiveness and efficiency of decision making processes, 2) data/knowledge representation, and 3) processing and presenting numerical data, text, images, animation, sound, and video. Capabilities found in traditional DSS and GDSS

(Group DSS) can be used for generating and evaluating more alternatives, do simulations and quantitative analyses. Many of a manager's decisions are single shot decisions where, for example, decision analysis tools can be used to get consistent decisions. Decision making technologies could also be used to support team meetings. GDSS, if properly supported by group process techniques [25], can be useful as MSS [73]. GDSS can have tools for: electronic brainstorming, idea organization, voting, stakeholder identification and analyses [53]. For a recent treatment of how firms can use business analytics, including decision support and business intelligence, to compete, see Davenport and Harris [20].

For the MSS-OS two capabilities are keys: 1) filtering and, 2) monitoring. Filtering techniques can be used to facilitate electronic communication applying artificial intelligence techniques to sort, distribute, prioritize, and automatically respond to electronic messages [66]. There is also monitoring and filtering techniques to be used for environmental scanning. These techniques include, for example, the use of intelligent agents and push technology for scanning the internet. Increasingly we can expect that Web 2.0 technologies will be used in MSS-OS, for example, having CEOs and other top-managers use blogs.

For the MSS-HR two capabilities are keys: 1) communication for fostering and supporting individuals and teams, and 2) coordination. Coordination technologies, like project management tools and calendars, can be used to manage and organize the execution of decisions and processes. A major purpose of MSS-HR is to help a manager communicate information that will motivate and allow organizational members to be creative within defined limits of freedom [65]. Communication technologies can be used to communicate: 1) basic values, purposes, and direction for organizational members, and 2) codes of business conduct and operational guidelines. For these purposes ICT like e-mail, videoconferencing, electronic documents, and multimedia can be useful. Although the media richness theory [19] suggests that a technology like e-mail is a lean medium (compared to face-to-face), but research suggests that lean media can be used in rich ways if an organization encourages and supports rich use [45].

The output of this phase will be a specification of what should be built.

#### **4.4 The Approach in Work**

The MSS design approach has been developed using current management and executive theory, and current MSS knowledge and models. It has in varying degrees been used in real applications (18 applications), in management programs, and it has been used in quasi-experimental lab studies. Over the years the approach has been enhanced.

We have in the design process used MSS exemplars. Exemplars are descriptions of how different types of MSS can be used to support different managerial roles. The focus in the exemplars is on use and impacts of use and less on technical aspects. The use of exemplars seems to be useful.

We have used Decision Explorer (from Banxia) and GroupSystems (from Ventana) in the second phase. Decision Explorer has been used to identify and present "misfits" and to present and discuss remedies. GroupSystems has been used in lab studies to work through the first three phases, with an emphasis on the second and third phase. Both ways to use computer-based

support seem to enhance the process and they both seem to be worthwhile to further explore.

Research suggests that there are changes in the criteria of effectiveness over an organizational life cycle [55]. Another emphasis change can be found with regard to management level [54]. Implications of these findings are that the importance and criticality of effectiveness criteria and managerial roles will vary over time as well as between managerial levels. This knowledge has been used when design MSS.

## **5. CONCLUSIONS AND FURTHER RESEARCH**

Although, we see a change in what can be modeled using traditional modeling approaches (primarily process modeling)—for example, Seidel et al. [64] show how a creative process in the screen business can be modeled—the nature of managerial work [50, 51] makes it impossible to use traditional modeling approaches. The MSS design approach has been presented as a new guide for MSS design and development. The content of the approach builds on Quinn and associates' competing values model and current MSS knowledge. The process of the approach builds on prescriptions found in most MSS writings (for good summaries see Hartono et al. [35] and Clark et al. [17], for example, regarding the relationship between the MSS designers and the users (managers), the importance of having sponsors and champions, the iterative process, and the use of prototypes and exemplars.

Future research on the MSS design approach will include the development of better instruments and better support for the second and third phase (this includes the development of computer-based support). Future research will also include empirical studies addressing the relationship between MSS use and support for managerial cognition, decision, and action and how this is linked to individual and organizational performance. The result can improve our ability to design MSS and prescribe how MSS can be used to improve individual and organizational effectiveness.

Recently, Cameron [6] presented a fifth model, the multiple constituencies model, having a customer focus, being boundary spanning and addressing competitiveness. Cameron, as well as other researchers, advocates Positive Organizational Scholarship (POS) which focuses on positive deviance and extraordinary performance that stretch beyond the traditional levels of performance labeled as effective. We are currently trying to incorporate the "new" model as well as "extended" views on effectiveness in our MSS approach.

The design science process for developing the approach was at the beginning quite eclectic. At the beginning of the process no prescriptive literature existed on how to do design science research. Later we were influenced by Walls et al. [71] and especially their view on the use of kernel theories. Currently we are focusing on how to "package" the approach and related knowledge in such a way that it can be useful to practitioners. In this work we are influenced by the work of Carlsson [9] and Hrstinski et al. [37].

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